

From spill events to catchment-scale prediction

Environmental drivers and predictability of combined sewer overflow duration behaviour across England

Project focus. A national analysis linking event-level CSO monitoring records to wastewater treatment works (WWTW) catchments, fitted long-duration tail parameters, environmental characteristics, adjusted statistical models and out-of-sample prediction.

Author

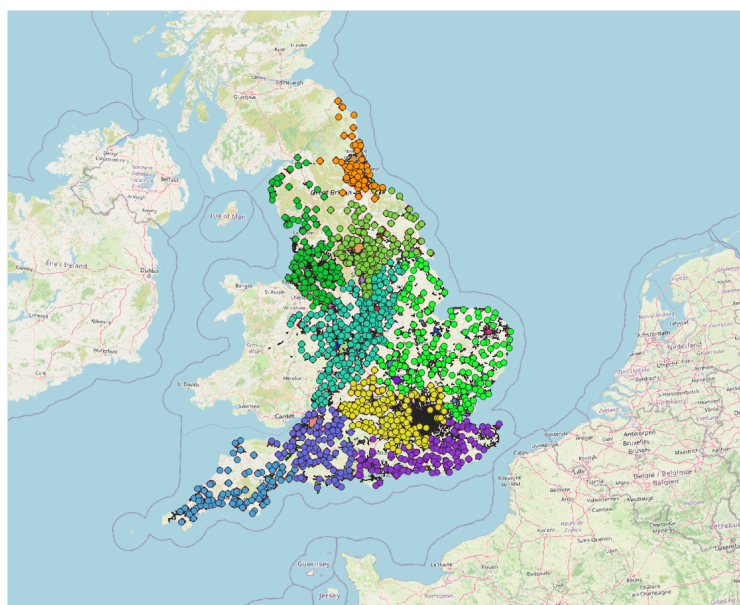
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Research setting

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This report follows the logic of a technical research paper: problem, data, parameterisation, spatial linkage, explanatory analysis, adjusted inference, predictive validation, synthesis and limitations. The visual language is intentionally restrained so the scientific argument carries the report.

Abstract

Combined sewer overflows (CSOs) are usually summarised by how often they spill or how long they spill in total. Those summaries do not fully describe the persistence of the longest events. This project constructs a national event-level workflow that begins with water-company Event Duration Monitoring records and ends with four related but distinct outcomes at WWTW-associated sensors: the fitted long-duration tail-shape parameter β , the fitted inverse-timescale parameter λ , mean spill duration, and annualised spill frequency. Events lasting more than 240 minutes define the fitted long-duration tail. Sensor records are linked to WWTW sewersheds, treatment capacity and a common external predictor framework containing rainfall, hydrogeology, land cover, population, infrastructure-age proxy, terrain and catchment scale. The analysis deliberately progresses from unadjusted one-variable associations to multivariable adjustment and finally to leakage-safe supervised regression evaluated on unseen WWTWs. The central scientific question is not simply which variables correlate with CSO behaviour, but how much of the spatial variation can be explained and predicted from measured external catchment characteristics. The final predictive programme finds modest rather than decisive out-of-sample skill, which is itself informative: static catchment descriptors contain structure, but important event-scale hydraulic and temporal information remains unobserved.

Executive summary

THE QUESTION

Can the environmental and operational characteristics of a WWTW catchment explain why its associated CSO sensor has a particular long-duration spill distribution, and can those characteristics predict behaviour at an unseen WWTW?

THE ANALYTICAL STORY

The project first created a common national event dataset, then quantified the long-duration tail using β and λ . Those sensor-level outcomes were made interpretable spatially by linking sensors to WWTW sewersheds. External catchment variables were then extracted at the same spatial unit. One-variable regressions identified candidate relationships, multivariable regression was designed to determine which relationships survived adjustment, and the final machine-learning stage asked whether the resulting information could generalise to unseen WWTWs.

Table 1: Current final predictive scorecard. RMSE values are on the logged model scale and are only comparable within a target.

Target	Selected model	Test RMSE	Dummy RMSE	Test R^2	Interpretation
β	OLS, interpretable static	0.653	0.687	0.073	4.9% lower RMSE; modest signal, bootstrap improvement still includes zero.
λ	OLS, interpretable static	3.855	4.056	0.095	4.9% lower RMSE; weak and inconclusive because improvement uncertainty includes zero.
Mean duration	Ridge, static + annual rainfall	1.631	1.675	0.052	2.6% lower RMSE; modest signal, with substantial extra value from company context.
Annualised frequency	Ridge, static + annual rainfall	1.341	1.442	0.133	7.0% lower RMSE; clearest baseline improvement of the four new models.

THE CENTRAL CONCLUSION

Measured spatial context is informative but not sufficient for precise site-level prediction. Population and hydrogeology carry signal for the fitted tail parameters, rainfall contributes to duration, and intergranular-flow geology plus annual dry-spell information contribute to annualised frequency. Yet the learning curves are already close to a plateau. The next major gain is therefore more likely to come from richer hydraulic and time-resolved process information than from fitting a more complicated algorithm to the same static variables.

Contents

Abstract	i
Executive summary	i
1 Introduction	1
1.1 Why long-duration CSO behaviour matters	1
1.2 The four outcomes	1
1.3 Research questions	1
1.4 Contribution and scope	1
2 From water-company records to an authoritative sensor master	1
2.1 Data acquisition	1
2.2 Cleaning and standardisation	2
2.3 Sensor identity	2
3 The long-duration spill distribution	2
3.1 Why focus on events longer than four hours?	3
3.2 Stretched-exponential model	3
3.3 Sensor-level fitting and uncertainty	3
3.4 National parameter distributions	4
4 Linking CSO sensors to WWTW catchments	5
4.1 Wastewater catchments and WWTW sewersheds	5
4.2 Coordinate-based containment baseline	5
4.3 Why a second WWTW-associated-sensor strategy was needed	5
4.4 Place-core and supporting-evidence strategy	5
5 Building the external catchment predictor matrix	5
5.1 Treatment capacity	7
5.2 Hydrogeology	7
5.3 Rainfall	7
5.4 Urban form, population, age, terrain and scale	7
6 One-variable associations across four outcomes	8
6.1 Method	9
6.2 Tail shape β	9
6.3 Inverse timescale λ	9
6.4 Mean spill duration and annualised frequency	9
6.5 Cross-outcome synthesis	9
7 Multivariable regression: which relationships survive adjustment?	11
7.1 Why adjustment is necessary	11
7.2 Pre-specified model ladder	11
7.3 Encoding correlated and compositional predictors	11
7.4 Adjusted beta findings	11
7.5 Adjusted lambda, duration and frequency findings	11
7.6 Diagnostics and spatial structure	12
8 Predicting unseen WWTWs with multivariable regression models	12
8.1 Four separate supervised prediction problems	12
8.2 The 70/15/15 protocol	12
8.3 Primary performance metric	13
8.4 Final model results	13
8.5 Did the annual NIMROD rainfall block add value?	14
8.6 Which measured variables does the final model rely on?	14
8.7 Is the project mainly limited by sample size?	14

9 Integrated discussion	16
9.1 From association to mechanism	16
9.2 Why beta, lambda, duration and frequency must remain separate	16
9.3 Explanation is not prediction	16
9.4 What the machine-learning result adds to the scientific story	16
10 Limitations	16
11 Conclusions and next research steps	17
11.1 What this project established	17
11.2 Priority next steps	17
A Data provenance and company coverage	18
B Full variable dictionary	18
C Complete univariate result atlas	18
D Multivariable model specifications and diagnostics	18
E Machine-learning technical appendix	18
F Reproducibility	18
References	19

1 Introduction

1.1 Why long-duration CSO behaviour matters

A combined sewer overflow is a pressure-relief mechanism in a sewer network. During sufficiently high hydraulic loading, diluted sewage and stormwater can discharge to the environment rather than continuing through the network and treatment process. Public reporting often focuses on event counts or annual duration. Those quantities are important, but two sites with similar counts can still have very different distributions of event duration. One may experience many relatively contained events, while another has a much more persistent upper tail.

This project therefore asks a different question: *what controls the shape and timescale of the longest spills, and how predictable are those characteristics from the external environment of the wastewater catchment?*

1.2 The four outcomes

The final analysis deliberately separates four quantities that answer different physical questions.

Table 2: Four outcomes used throughout the report.

Outcome	What it measures	Interpretation
Tail shape β	Shape of the duration distribution conditional on an event lasting > 240 min	Lower β means a heavier, more persistent long-duration tail; higher β means a faster-decaying tail.
Inverse timescale λ	Fitted scale component in $S(t) = \exp[-(\lambda t)^\beta]$	At fixed β , higher λ means a shorter characteristic time $1/\lambda$. It is not conventional Weibull η .
Mean spill duration	Observed mean duration of valid events	Direct site-level duration burden, expressed in minutes before transformation.
Annualised spill frequency	Valid event count divided by documented monitoring years	Exposure-adjusted frequency; preferable to raw lifetime count when monitoring periods differ.

1.3 Research questions

The report is organised around five linked questions:

1. What does the national event-level CSO duration distribution look like, particularly above four hours?
2. Can stable sensor-specific β and λ values be estimated from those long-duration events?
3. Can those sensor outcomes be linked credibly to WWTW catchments and external catchment characteristics?
4. Which external variables show isolated and then adjusted associations with β , λ , duration and frequency?
5. Given all available external information together, how accurately can each outcome be predicted at an unseen WWTW?

1.4 Contribution and scope

The project combines three methodological ideas: event-level EDM data rather than only annual summaries; sensor-level long-duration distribution parameters rather than a single national distribution; and a common geospatial predictor matrix linked to WWTW sewersheds. The study is observational. Statistical or predictive association is evidence of structure, not proof of a causal hydraulic pathway.

The research question can only be answered if the event data from different water companies are first placed on a common, auditable footing. The next section therefore starts with acquisition and harmonisation rather than modelling.

2 From water-company records to an authoritative sensor master

2.1 Data acquisition

Event-level spill records were assembled across the participating English water companies using supplied EDM files and Environmental Information Regulation requests where required. The common analytical record contains a location identifier, permit or asset identifier, event start time, event end time and duration in minutes.

[TO COMPLETE: Insert a concise company-by-company data source and year-coverage table. Do not narrate every email or download. Summarise only differences that affect comparability.]

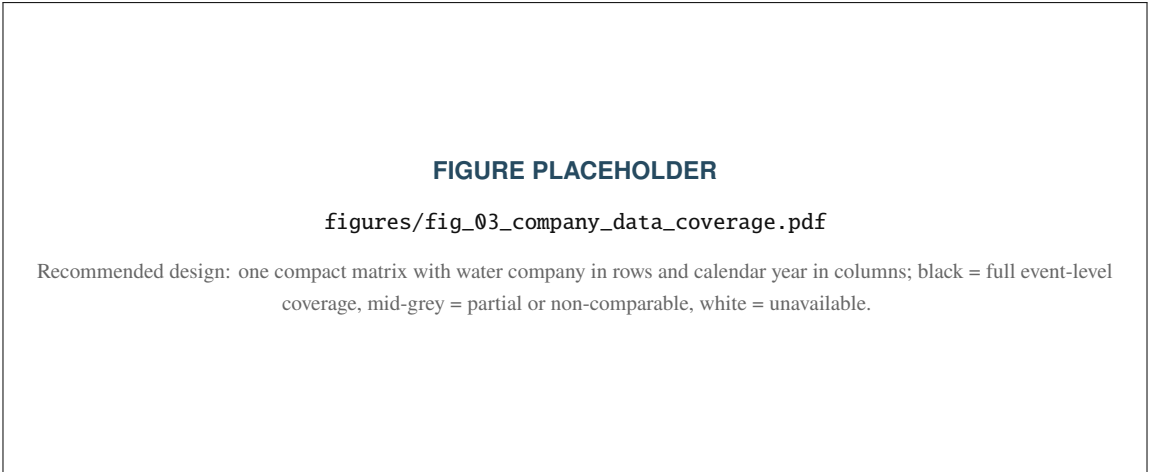


Figure 1: National EDM data coverage used to construct the event-level analysis.

2.2 Cleaning and standardisation

Company-specific files were standardised into one event schema. Duration was calculated or checked in minutes from start and stop timestamps, with impossible or ambiguous rows retained in an audit path rather than silently coerced. This step was important because company formats and conventions differ, and errors in units or timestamp construction would propagate directly into the fitted duration distribution.

2.3 Sensor identity

Events were grouped into an authoritative sensor identity, preserving company, location name and permit or asset identifier. The national sensor master retained 25,409 unique sensors, of which 11,561 had accepted coordinates and 8,746 had a finite fitted β in the earlier national fitting release. A sensor remained in the master even if coordinates or a fitted tail parameter were unavailable, so missingness remained visible rather than becoming an invisible sample restriction.

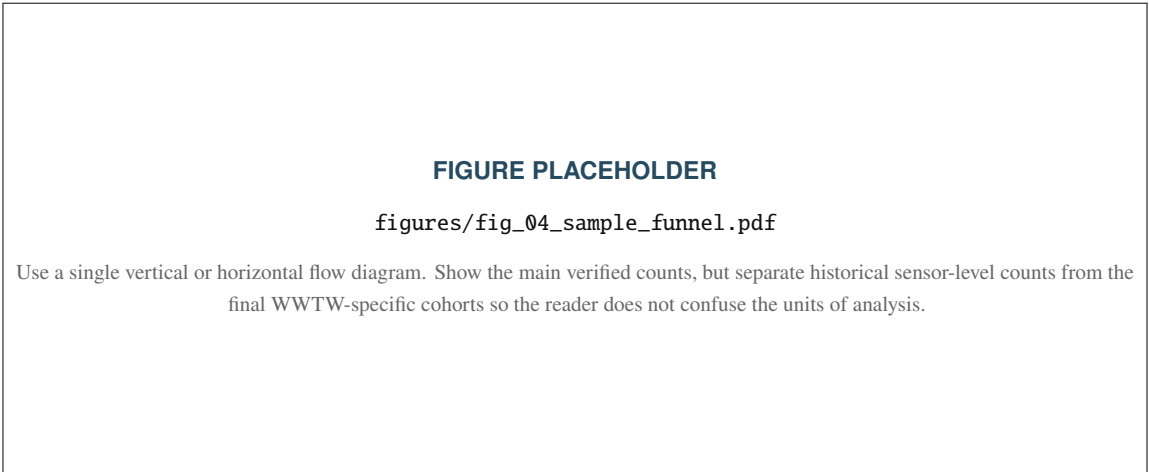


Figure 2: Project sample funnel from raw sensor identities to the final WWTW-level modelling cohorts.

Once the event records are comparable, the project can move from administrative records to a physical statistical object: the distribution of spill durations at each sensor.

3 The long-duration spill distribution

3.1 Why focus on events longer than four hours?

The active tail definition is fixed as

$$T > 240 \text{ minutes.} \quad (1)$$

The threshold separates the long-duration regime used for tail fitting from the complete event distribution. It is a design choice and must remain constant throughout the report.

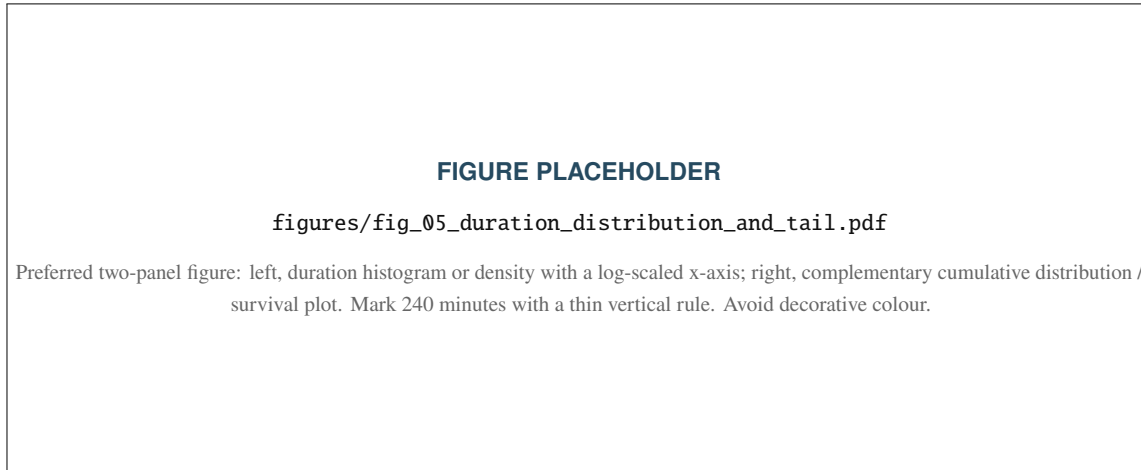


Figure 3: National spill-duration distribution and the long-duration analysis threshold.

3.2 Stretched-exponential model

For long-duration events, the fitted survival function is

$$S(t) = P(T > t) = \exp [-(\lambda t)^\beta], \quad (2)$$

where β controls tail shape and λ is an inverse timescale. At fixed λ , lower β produces a heavier, more persistent upper tail. At fixed β , higher λ shortens the characteristic timescale $1/\lambda$.

$$\beta < 1 \Rightarrow \text{stretched-exponential, persistent tail,} \quad \beta = 1 \Rightarrow \text{exponential,} \quad \beta > 1 \Rightarrow \text{compressed, faster-decaying tail.} \quad (3)$$

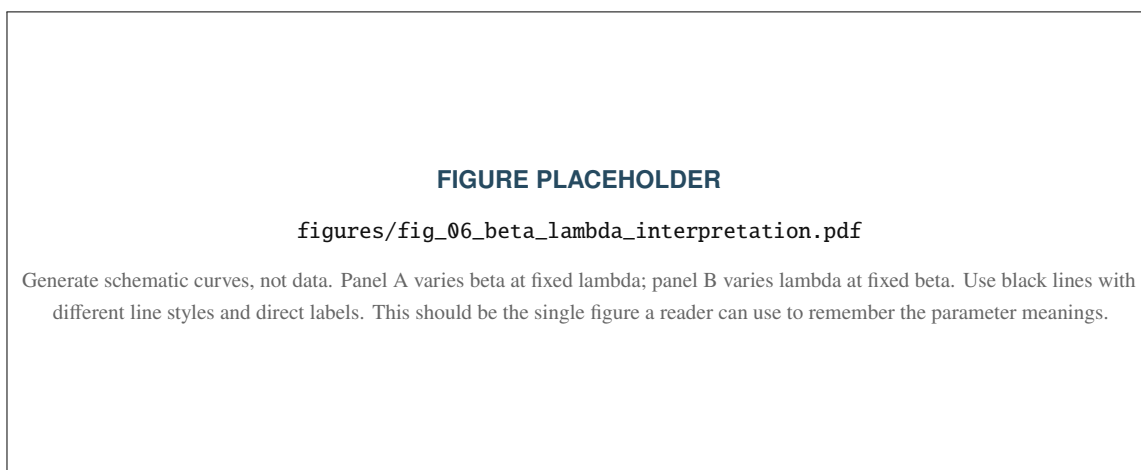


Figure 4: How β and λ change the fitted survival curve.

3.3 Sensor-level fitting and uncertainty

For each sensor with sufficient long-duration observations, the tail was fitted independently. Bootstrap resampling of observed tail events was used to describe parameter stability and to distinguish higher-quality estimates from broader finite fits. Bootstrap precision is not proof that the stretched-exponential model is a perfect physical representation; it describes uncertainty under the fitted empirical sample.

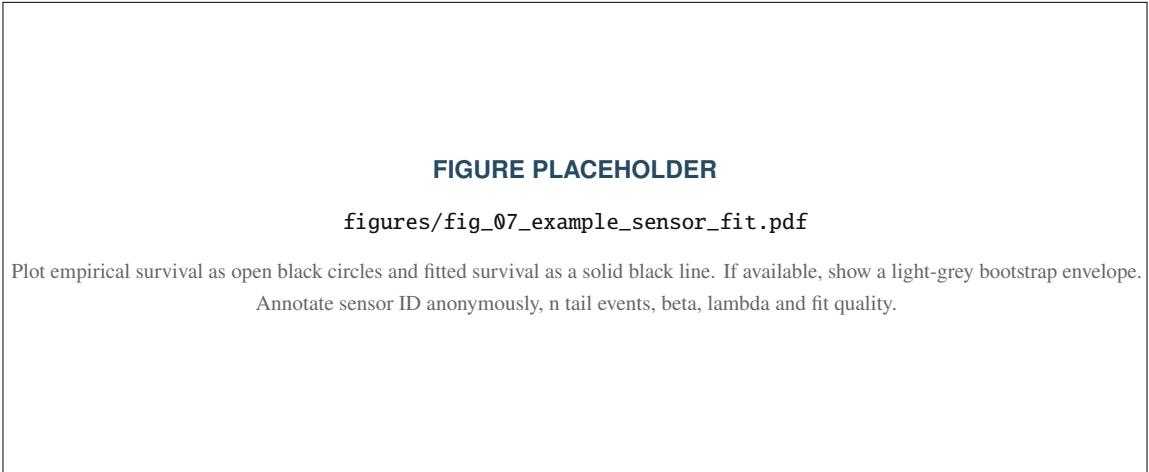


Figure 5: Example empirical long-duration survival curve and fitted stretched-exponential model for one representative sensor.

3.4 National parameter distributions

The broad WWTW-linked beta cohort contains 819 finite positive values in the current analysis, while the strict-quality sensitivity contains 227. The broad median beta is approximately 0.554 and the strict median approximately 0.355, placing the central tendency of both samples below the exponential reference $\beta = 1$.

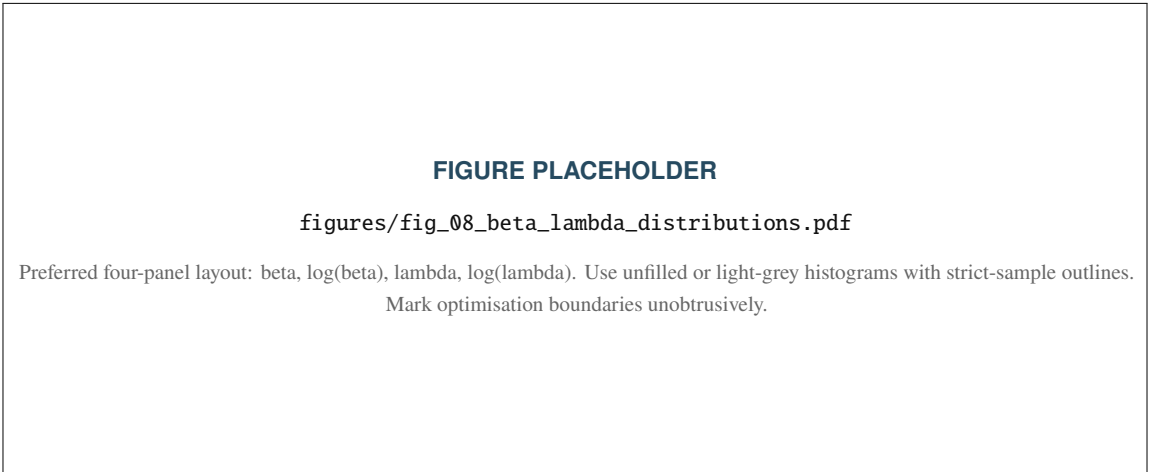


Figure 6: Distributions of fitted β and λ on modelling scales.

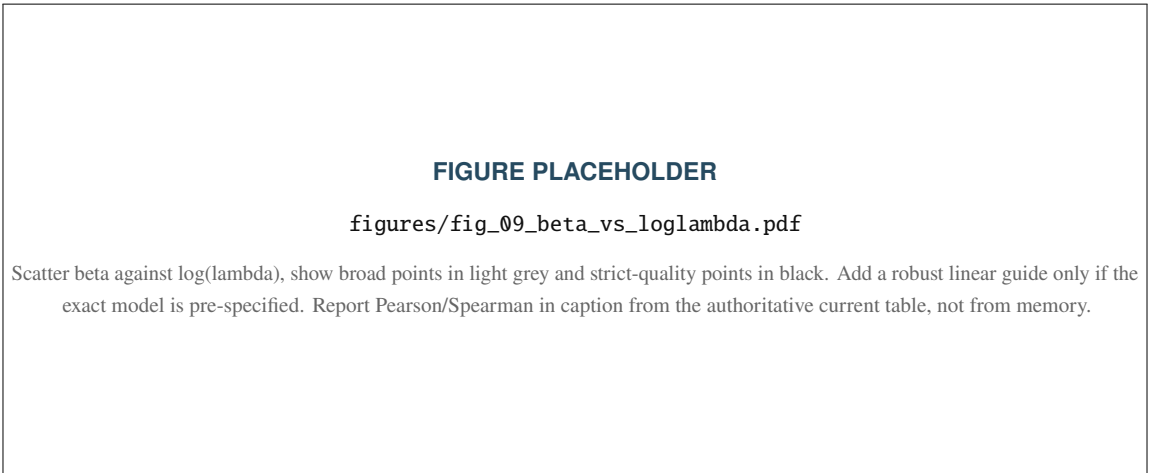


Figure 7: Joint relationship between fitted tail shape and inverse timescale.

The fitted parameters now exist at individual sensors, but the candidate explanations - rainfall, geology, population, capacity and land cover - exist at catchment or WWTW scale. The next challenge is therefore spatial identity: which sensor belongs to which wastewater system?

4 Linking CSO sensors to WWTW catchments

4.1 Wastewater catchments and WWTW sewersheds

Published wastewater catchment polygons were linked to treatment-work identifiers and dissolved by company and WWTW code to form system-scale sewersheds. These polygons provide the spatial unit over which capacity and environmental characteristics are attached.

4.2 Coordinate-based containment baseline

The first national assignment compared coordinate-eligible sensors in British National Grid (EPSG:27700) with same-company wastewater catchment polygons. Of 11,561 coordinate-eligible sensors, 8,960 were accepted through unique same-company containment or overlapping polygons that resolved to the same WWTW, an acceptance rate of approximately 77.5%.

4.3 Why a second WWTW-associated-sensor strategy was needed

The supervisor’s revised question focuses on overflow sensors plausibly associated directly with treatment works. A first fuzzy full-string name matcher was deliberately conservative but had low recall. Diagnostic review showed that facility suffixes such as “SSO”, “CSO”, “STW”, “WRC”, “storm tank” and “outfall” could dominate a similarity score even when the geographic place core was identical.

4.4 Place-core and supporting-evidence strategy

The refined strategy therefore separates geographic identity from facility terminology. For example, “SCAYNES HILL SSO” and “SCAYNES HILL STW” reduce to the shared place core “SCAYNES HILL”. Automatic matching remains same-company and beta-blind, preserves directional qualifiers such as NORTH and SOUTH, uses score-margin and ambiguity rules, and may use existing WWTW assignment, distance or regulatory outlet information as supporting evidence. Proximity alone is not treated as hydraulic proof.

Table 3: Evidence hierarchy for the final sensor-WWTW linkage.

Tier	Evidence
Regulatory-confirmed	Genuine regulatory storm-overflow or storm-tank identity linked to the WWTW, same company and no unresolved conflict.
Convergent high confidence	Strong place-core evidence plus one or more independent supports such as existing WWTW agreement, close outlet/site distance, sewershed consistency or a strong candidate margin.
Manually confirmed	Plausible but incomplete cases reviewed explicitly without inspecting beta or environmental outcomes.
Excluded / unresolved	Proximity-only candidates, cross-company conflicts, final-effluent-only sites or unresolved multiple-WWTW cases.

EVIDENCE NOTE Published catchment containment, names and public coordinates do not reconstruct the underground sewer network. Pumping, diversions and historical connections may be absent. The spatial assignment is a defensible analytical linkage, not a claim of exact pipe connectivity.

A defensible spatial link converts each fitted sensor outcome into a catchment-scale research unit. Only at this point is it meaningful to ask whether geology, rainfall, urban form or capacity correspond to systematic differences in spill behaviour.

5 Building the external catchment predictor matrix

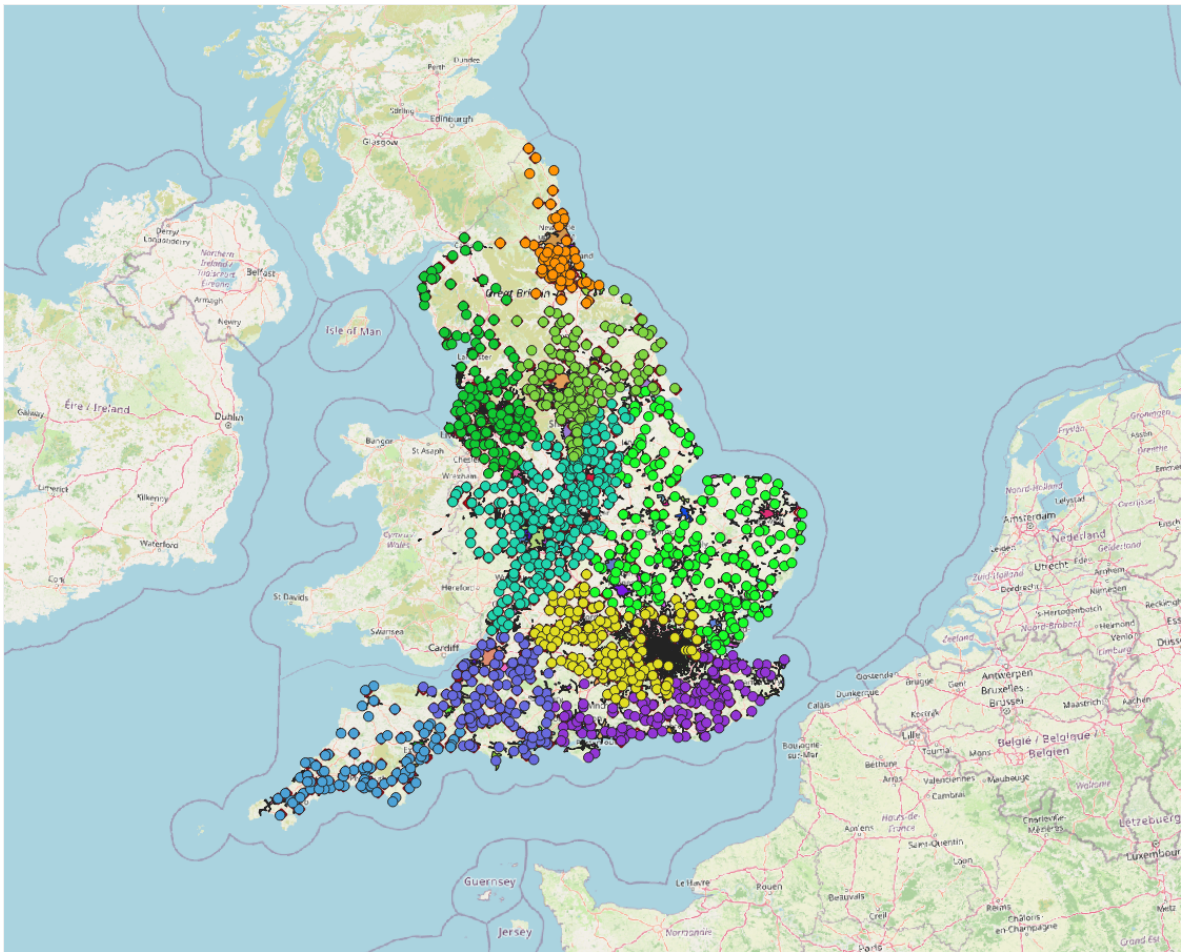


Figure 8: National spatial overview of WWTW catchments and associated CSO sensors.

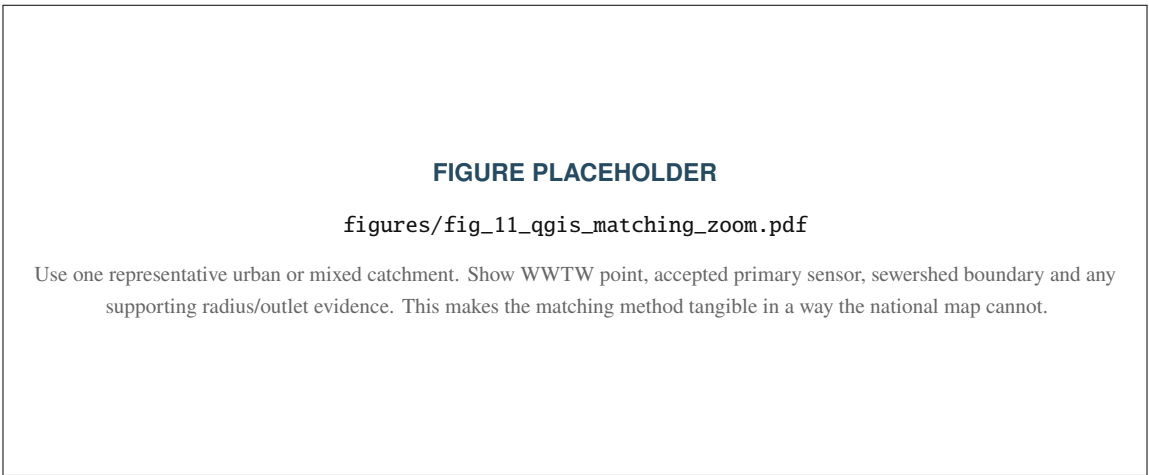


Figure 9: Example of the final sensor-to-WWTW linkage at local scale.

5.1 Treatment capacity

Treatment capacity utilisation is defined as

$$R_{\text{capacity}} = \frac{\text{load entering the WWTW}}{\text{reported design capacity}}. \quad (4)$$

The historic distribution has a median near 0.935, with many values exactly equal to 1 and none above 1. This may reflect reporting or censoring and is treated cautiously rather than as an exact wet-weather hydraulic headroom measure.

5.2 Hydrogeology

BGS Hydrogeology 625K polygons were intersected with each WWTW sewershed. For hydrogeological category k and sewershed j ,

$$H_{jk}(\%) = 100 \frac{A(C_j \cap G_k)}{A(C_j)}. \quad (5)$$

The percentage representation captures catchment composition rather than assigning a single point geology. Productivity shares and flow-mechanism shares are compositional, so exact redundant categories are not entered simultaneously in ordinary regression.

5.3 Rainfall

Two complementary rainfall descriptions are retained. Long-term climatic context is represented by static mean annual and winter rainfall. Monitoring-period variability is represented by annual NIMROD indices such as total wet-day precipitation, maximum 1-day and 5-day totals, rainfall intensity, heavy-rain days, consecutive wet days and consecutive dry days. Annual indices are area-weighted over the WWTW sewershed and require comparable monitoring-period coverage.

5.4 Urban form, population, age, terrain and scale

Land cover, population density, pre-1973 building share, mean terrain slope and WWTW sewershed area represent complementary aspects of runoff connectivity, wastewater loading, infrastructure age proxy, topographic response and system scale.

Table 4: Core external variables used in explanatory and predictive modelling. Extend this table only when a variable is scientifically distinct.

Variable family	Example field / representation	Unit	Why it may matter
Hydro productivity	high, moderate, low, no-groundwater shares	% area	Groundwater storage and transmission may alter infiltration/inflow persistence.
Hydro flow mechanism	intergranular, fracture-flow shares	% area	Distributed porous storage and preferential fracture pathways imply different hydraulic response timescales.
Static rainfall	mean annual; mean winter rainfall	mm	Long-term wetness, recharge and seasonal saturation context.
Annual NIMROD rainfall	PRCPTOT, Rx1day, Rx5day, SDII, R10, R20, CWD, CDD	mm / days	Monitoring-period rainfall amount, intensity and persistence.
Land cover	urban, suburban, continuous developed-land index	% / index	Imperviousness, connected drainage and urban hydraulic complexity.
Population	population density	people km ⁻²	Proxy for wastewater loading and urban system scale; not a direct causal mechanism by itself.
Infrastructure age proxy	pre-1973 building share	%	Approximate spatial proxy for older sewer infrastructure where direct sewer age is unavailable.

Variable family	Example field / representation	Unit	Why it may matter
Terrain	mean slope	degrees	Affects runoff concentration, routing and pumping context.
Treatment operation	capacity utilisation ratio	ratio	Approximate loading relative to reported design capacity.
System scale	WWTW sewershed area	km ²	Routing distance, spatial averaging, storage and asynchronous subcatchment response.
Company context	water company category	category	Captures operator/region context; used as adjustment/sensitivity, not treated as a physical environmental variable.

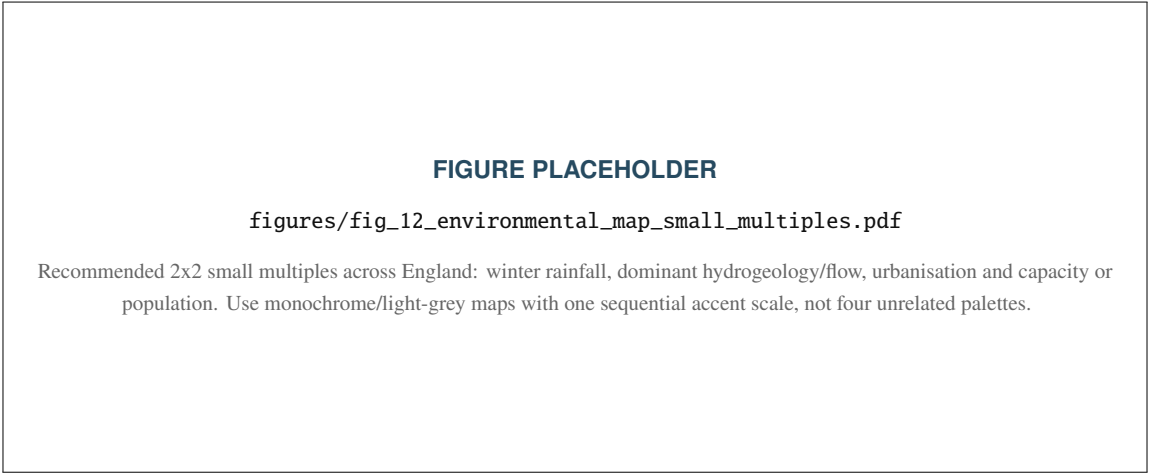


Figure 10: Examples of spatial variation in the external predictor matrix.

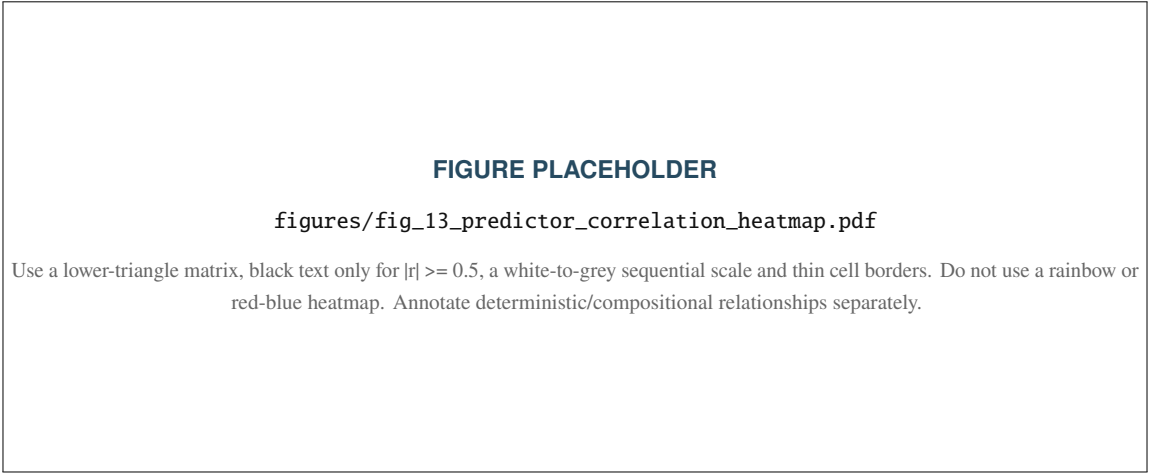


Figure 11: Correlation structure among the final external predictors.

The common predictor matrix now allows an initial question: what does each external variable appear to do on its own? That is useful for identifying patterns, but it cannot separate correlated mechanisms.

6 One-variable associations across four outcomes

6.1 Method

For a positive outcome Y_i , one-variable models use

$$\log(Y_i) = \alpha + bX_i + \epsilon_i, \quad (6)$$

with HC3 robust covariance. Effects are reported as exact proportional differences for a meaningful predictor increment ΔX ,

$$100 [\exp(b\Delta X) - 1] \%. \quad (7)$$

Benjamini-Hochberg false-discovery-rate adjustment is applied within the pre-specified predictor families. These are unadjusted associations, not causal effects.

6.2 Tail shape β

The strongest unadjusted beta relationships are associated with urban-system structure and selected hydrogeological categories. In the strict-quality sample, a doubling of population density corresponds to approximately 23.6% higher beta, while a 10-percentage-point increase in urban land cover corresponds to approximately 23.1% higher beta. Low-productivity and fracture-flow hydrogeology correspond to lower beta, whereas intergranular-flow hydrogeology corresponds to higher beta. Static annual and winter rainfall show negative beta associations in the strict sample, consistent with a heavier long-duration tail in wetter settings, but those rainfall signals weaken in the broader sample.

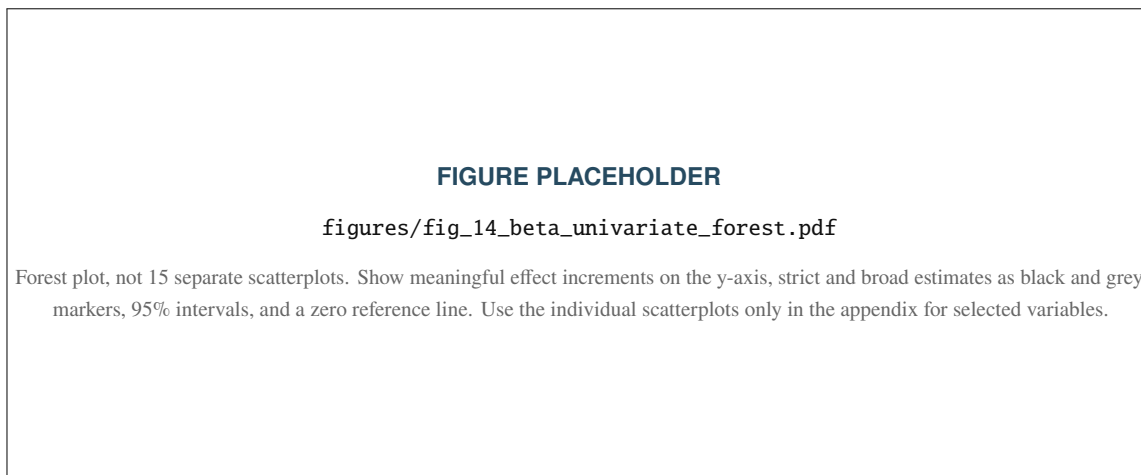


Figure 12: Unadjusted effects on fitted β .

6.3 Inverse timescale λ

The lambda analysis reveals a useful complement to beta. Low-productivity and fracture-flow settings tend toward higher lambda, while moderate-productivity and intergranular settings tend toward lower lambda in the strict analysis. Urbanisation and population density are associated with substantially lower lambda. Because beta and lambda are jointly estimated and lambda can be extremely variable, these results should be interpreted as parameter regimes rather than as isolated deterministic effects.

6.4 Mean spill duration and annualised frequency

Mean spill duration is more strongly structured by the measured external covariates than annualised frequency in the univariate stage, although even the largest single-predictor R^2 values are small. Suburban land cover and treatment-capacity utilisation show notable positive duration associations. Static winter rainfall and sustained multi-day annual rainfall also correspond to longer mean duration, while longer consecutive dry spells correspond to shorter duration. Frequency shows clearer hydrogeological and catchment-scale structure; intergranular-flow share and larger catchment area are associated with lower annualised frequency, and no annual NIMROD rainfall index survives the rainfall-family FDR screen.

6.5 Cross-outcome synthesis

The one-variable stage is scientifically useful because it shows that the four outcomes are not interchangeable. Variables that correspond to a heavier tail do not necessarily produce longer mean events or greater frequency. It also reveals the main problem that motivates adjustment: rainfall variables overlap with each other, urbanisation overlaps population and catchment scale, and hydrogeology percentages are compositional.

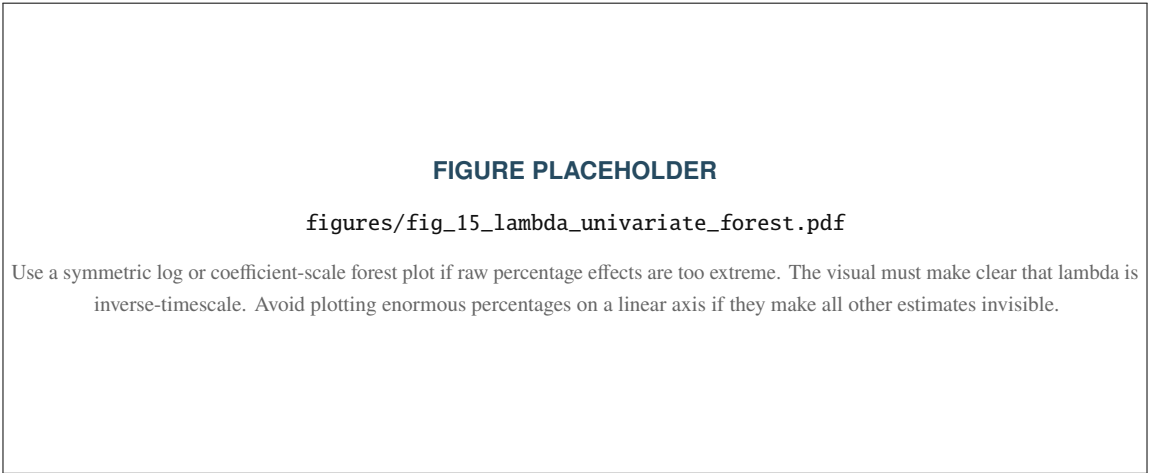


Figure 13: Unadjusted effects on fitted inverse-timescale λ .

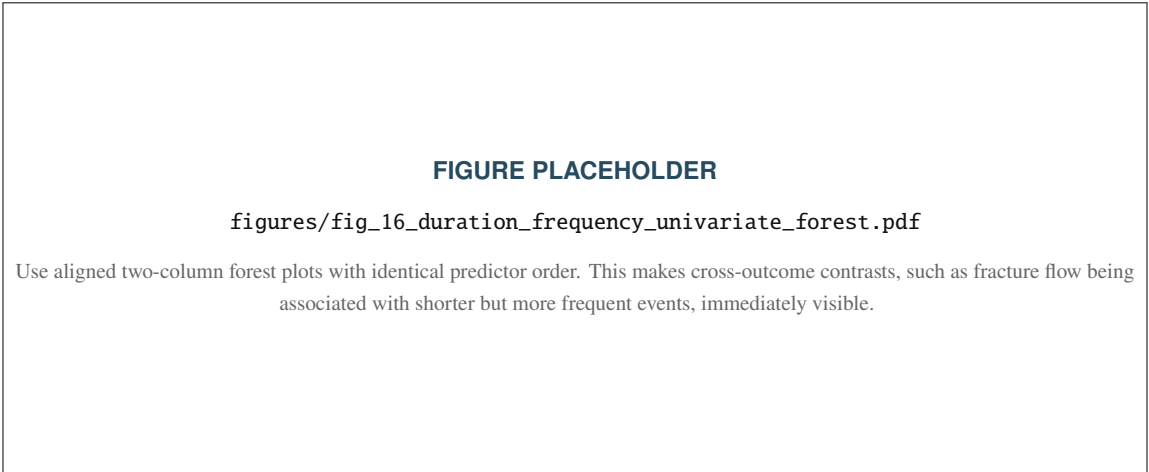


Figure 14: Unadjusted effects on mean spill duration and annualised spill frequency.

Table 5: Concise one-variable evidence synthesis. Keep the main text to the strongest patterns; place the complete coefficient atlas in the appendix.

Mechanism	Main observed pattern	Interpretation to carry forward
Urban system	Higher urban cover / population generally $\beta \uparrow$, $\lambda \downarrow$	Longer characteristic timescale but smoother/faster-decaying relative tail; strong overlap among urban proxies requires adjustment.
Hydrogeology	Intergranular/moderate and fracture/low-productivity often occupy opposite β - λ regimes	Flow architecture appears informative, but percentages are compositional and spatially structured.
Rainfall	Wetter winter climate tends toward $\beta \downarrow$; duration responds to winter and multi-day rainfall	Hydrological state and persistence may matter more than isolated intensity for long-duration behaviour.
System scale	Catchment area and capacity show outcome-specific associations	Operational and geometric scale can confound environmental patterns and must be adjusted jointly.

Univariate plots identify where the signal might be, but they cannot tell us whether rainfall still matters after geography, urbanisation and company are held constant. The next stage therefore moves from isolated associations to adjusted inference.

7 Multivariable regression: which relationships survive adjustment?

7.1 Why adjustment is necessary

A strong one-variable relationship can disappear after adjustment because another variable carries the same underlying information. Conversely, a weak one-variable relationship can strengthen once opposing confounding is removed. The purpose of the multivariable stage is therefore explanatory: estimate conditional associations, coefficient stability and uncertainty while keeping the model interpretable.

7.2 Pre-specified model ladder

The staged beta model follows the project roadmap rather than automated stepwise selection:

$$M0 : \log(\beta) \sim \text{company} + \text{capacity} + \log_2(\text{area}), \quad (8)$$

$$M1 : M0 + \text{hydrogeology}, \quad (9)$$

$$M2 : M1 + \text{compact rainfall representation}, \quad (10)$$

$$M3 : M2 + \text{urban/suburban cover} + \log_2(\text{population density}), \quad (11)$$

$$M4 : M3 + \text{slope} + \text{pre-1973 share}. \quad (12)$$

The final implementation should add the frozen annual dynamic-rainfall block only in a pre-specified extension, not by selecting rainfall indices after seeing the outcome.

7.3 Encoding correlated and compositional predictors

Hydrogeology productivity shares require a reference category or alternative compositional representation. Annual and winter static rainfall are near-redundant and should not simply be entered together in an interpretable model. Urban, suburban and continuous developed-land measures likewise require a deliberate representation. Variance inflation factors, coefficient paths and sensitivity models should be used to understand instability rather than mechanically deleting every correlated variable.

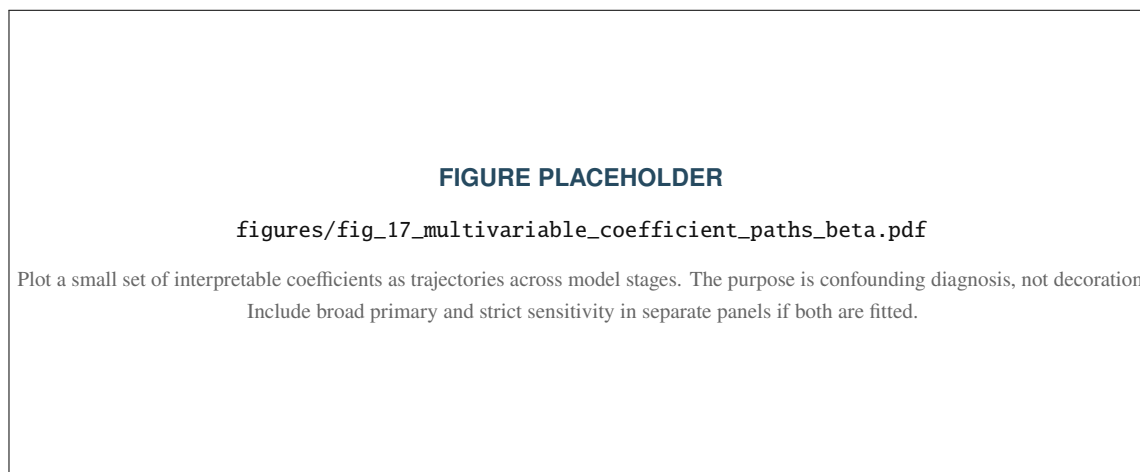


Figure 15: How beta coefficients change as adjustment blocks are added from M0 to M4/M5.

7.4 Adjusted beta findings

[TO COMPLETE: Insert final beta M0-M4/M5 results only after the multivariable release is frozen. For each major variable, state the adjusted magnitude, 95% interval, partial R², whether the coefficient attenuated/reversed, and whether it is stable in the strict-quality sensitivity.]

7.5 Adjusted lambda, duration and frequency findings

[TO COMPLETE: If final multivariable models are run for lambda, duration and frequency, summarise them here. Keep the main report balanced: beta/lambda are fitted tail parameters, while duration/frequency are

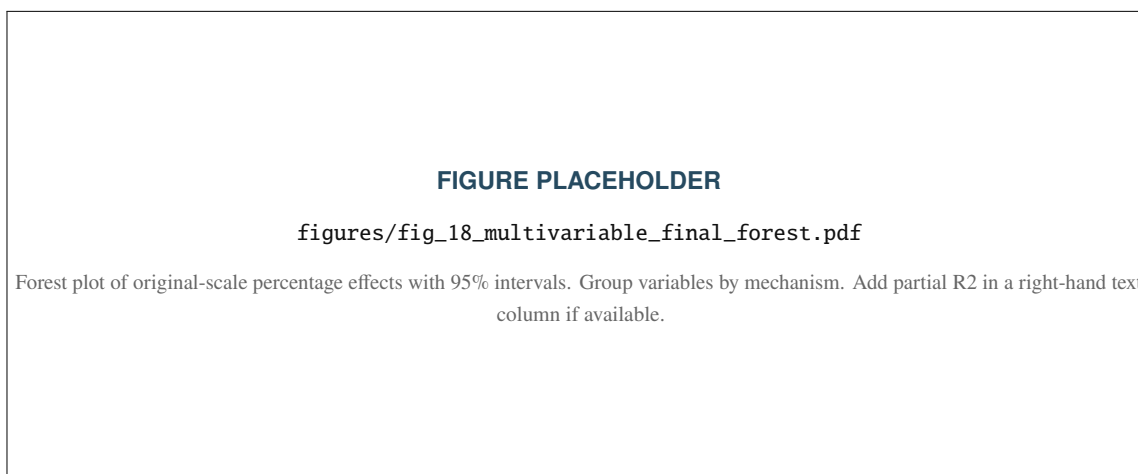


Figure 16: Final adjusted associations for the primary multivariable model.

observed outcomes. Do not use a large table of every coefficient in the body.]

7.6 Diagnostics and spatial structure

The final candidate model should be checked for nonlinear residual patterns, heteroskedasticity, influential WWTWs and spatial residual dependence. Moran's I or spatially grouped robustness is especially important because nearby WWTWs share climate, geology and company structure.

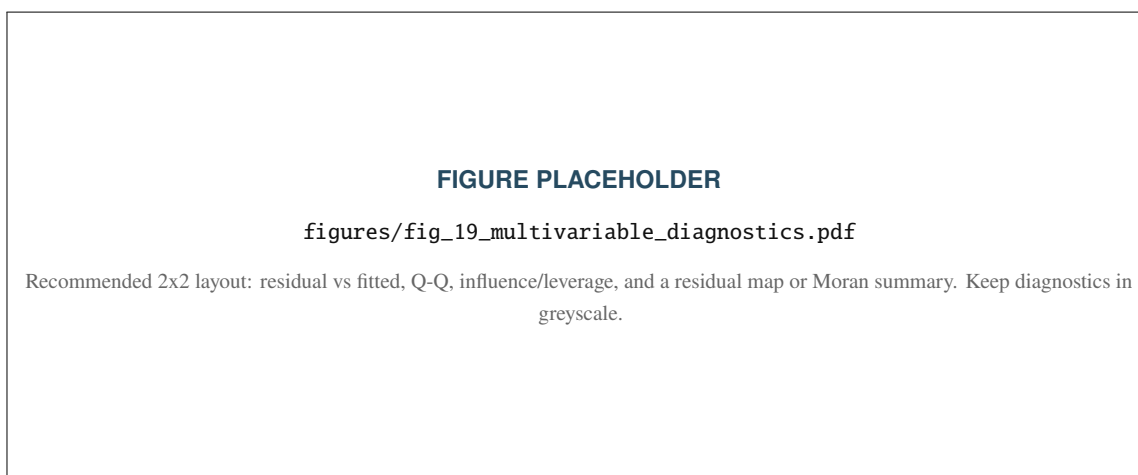


Figure 17: Diagnostics for the final adjusted model.

Adjusted regression asks whether a relationship remains after measured confounders are held fixed. That is not the same as prediction. A coefficient can be scientifically interesting even if the full model fails to forecast new WWTWs accurately. The final stage therefore changes the question from explanation to generalisation.

8 Predicting unseen WWTWs with multivariable regression models

8.1 Four separate supervised prediction problems

The predictive programme treats beta, lambda, mean duration and annualised frequency as four separate continuous targets. This preserves every WWTW with a valid label for the target in question rather than restricting all analyses to a common complete-case subset.

8.2 The 70/15/15 protocol

For each target, approximately 70% of WWTWs are used for training, 15% for validation and 15% for a locked test. Preprocessing, imputation and standardisation are fitted inside scikit-learn pipelines. Ridge and Elastic Net are tuned within the training portion; the validation set selects the frozen candidate; train plus validation are then refitted before

the test set is evaluated once.



Figure 18: Supervisor-directed prediction protocol. The locked test is not used to choose algorithms, predictors or hyperparameters.

8.3 Primary performance metric

The headline metric is root mean squared error,

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}. \quad (13)$$

Every model is compared with a training-fitted Dummy mean predictor. Improvement is

$$100 \frac{\text{RMSE}_{\text{Dummy}} - \text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{Dummy}}}. \quad (14)$$

A small positive improvement is not described as strong prediction simply because it is numerically below the baseline. Bootstrap uncertainty, R^2 , rank correlation and spatial/company robustness are considered alongside RMSE.

8.4 Final model results

The current final 70/15/15 release contains 819 beta WWTWs, 783 lambda WWTWs, 1,228 duration WWTWs and 967 annualised-frequency WWTWs. Beta and lambda select interpretable OLS specifications; duration and frequency select Ridge models using static plus annual dynamic rainfall information.

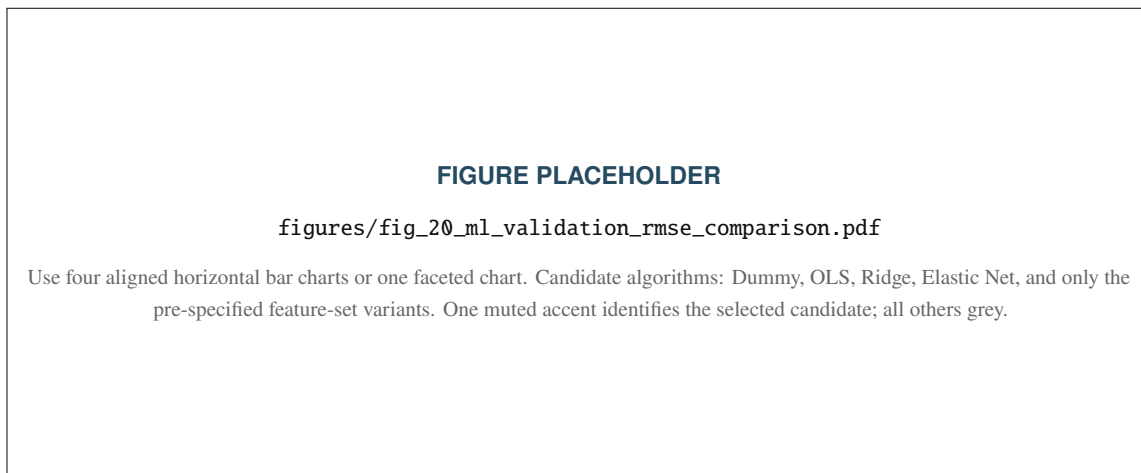


Figure 19: Validation RMSE used to select the final model for each target.

Table 6: Final 70/15/15 predictive results.

Target	Eligible n	Test n	Selected model	RMSE gain	R^2	Verdict
Beta	819	123	OLS, interpretable static	4.9%	0.073	Modest / qualified; improvement CI includes no gain.
Lambda	783	118	OLS, interpretable static	4.9%	0.095	Weak / inconclusive; parameter noise remains important.
Mean duration	1,228	185	Ridge, static + dynamic	2.6%	0.052	Modest / qualified; company context adds substantial information.
Annualised frequency	967	146	Ridge, static + dynamic	7.0%	0.133	Modest / qualified; bootstrap improvement remains above zero.

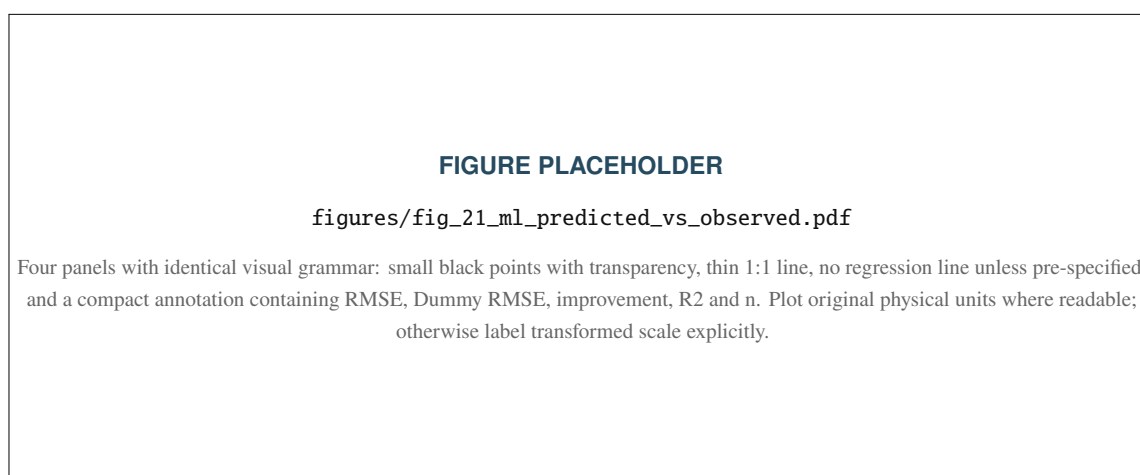


Figure 20: Locked-test predicted versus observed values for beta, lambda, mean duration and annualised frequency.

8.5 Did the annual NIMROD rainfall block add value?

The pre-specified static-versus-static-plus-dynamic comparison is more informative than asking whether a particular rainfall variable happens to be significant. Annual dynamic rainfall reduces test RMSE by approximately 0.015 for beta, 0.011 for duration and 0.054 for annualised frequency, while increasing scale-model RMSE by approximately 0.101.

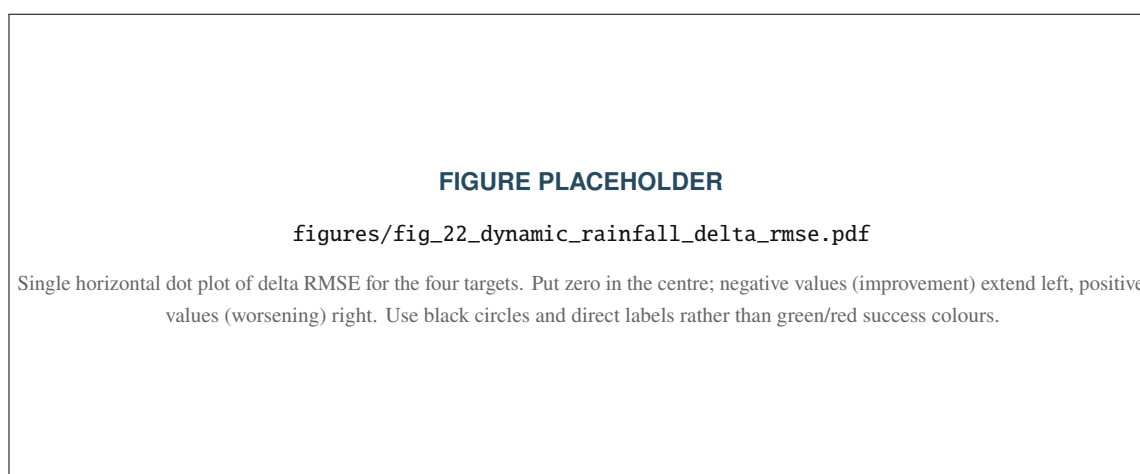


Figure 21: Value added by the annual NIMROD rainfall block.

8.6 Which measured variables does the final model rely on?

The most relied-on measured features differ by target. Population density, intergranular-flow geology and low-productivity geology are prominent for beta; the same families dominate scale. Duration relies most on static winter and annual rainfall plus annual SDII. Frequency relies on intergranular-flow geology, annual consecutive dry days and population density. These are predictive associations within correlated feature sets, not causal rankings.

8.7 Is the project mainly limited by sample size?

Learning curves for all four targets are close to a plateau at the available WWTW counts. This does not mean additional data are useless, but it argues against the idea that simply adding more similar WWTW rows will transform predictive accuracy. Richer process variables, target reliability and more temporally aligned rainfall/network information are more plausible next gains.

The final predictive stage puts the earlier results into perspective. The project has found real spatial structure, but the external variables explain only part of the site-to-site behaviour. The discussion therefore returns to the original physical question: what have we learned about the system, and what information is still missing?

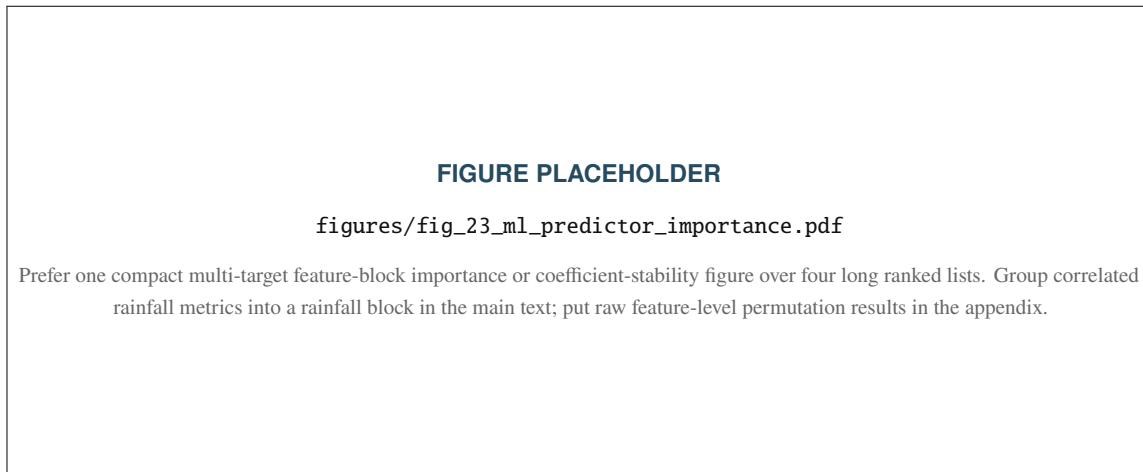


Figure 22: Stable predictor contributions in the final models.

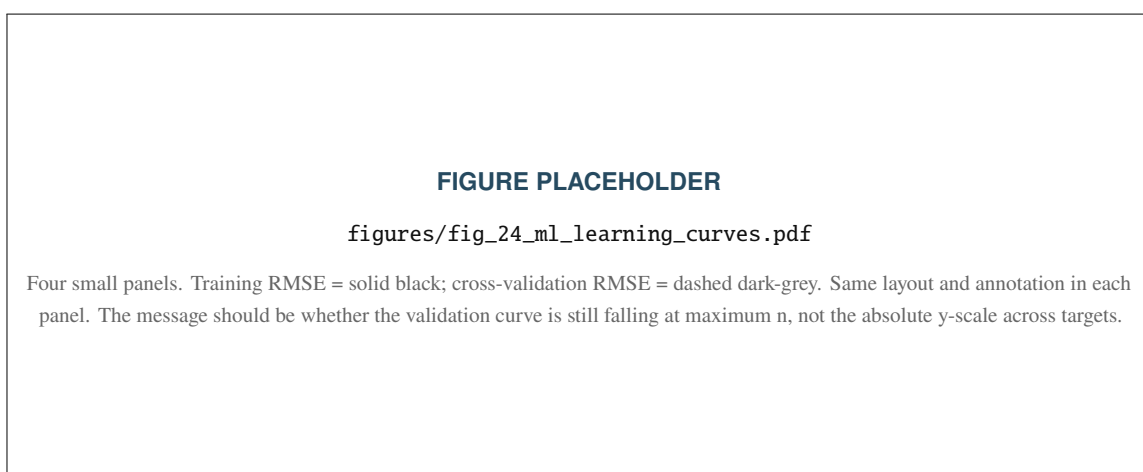


Figure 23: Learning curves for the four prediction targets.

9 Integrated discussion

9.1 From association to mechanism

The strongest result is not one universal environmental coefficient. It is the emergence of distinct response regimes. Urbanisation and population are associated with higher beta and lower lambda in the univariate parameter analysis, suggesting a longer characteristic timescale paired with a more regular relative tail. Intergranular and fracture-flow hydrogeology often occupy opposing beta-lambda regimes, which is more informative than treating aquifer productivity as a single monotonic scale. Rainfall appears more important for mean duration and for hydrological state than as a universal direct predictor of fitted tail parameters.

9.2 Why beta, lambda, duration and frequency must remain separate

The four outcomes encode different parts of spill behaviour. A catchment can have a heavy long-duration tail without high annual event frequency. A variable can alter the timescale of an event distribution without changing its relative tail shape in the same direction. The paired beta-lambda interpretation therefore prevents simplistic statements such as “higher rainfall means longer spills” when the evidence is actually about tail persistence.

9.3 Explanation is not prediction

Univariate and multivariable regression ask which relationships are present after specified adjustment. Machine learning asks whether the full predictor profile is sufficient to forecast an unseen WWTW. The modest test-set RMSE improvements show that these are not equivalent. A statistically clear association can exist while the model still leaves most site-to-site variation unexplained.

9.4 What the machine-learning result adds to the scientific story

A weak or modest model is not a failed final chapter. It quantifies the information ceiling of the measured feature set. The external environment matters, but the remaining prediction error points toward omitted hydraulic state: sewer topology, storage, pumping, maintenance, groundwater level, antecedent wetness, storm timing, treatment control and other time-varying operational conditions.

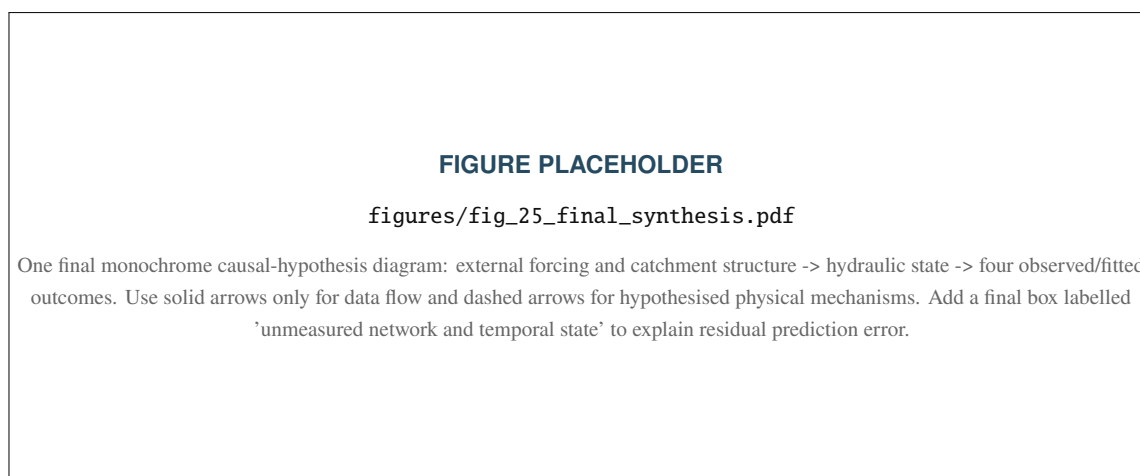


Figure 24: Integrated conceptual synthesis of the project.

10 Limitations

1. **Hydraulic connectivity is inferred, not observed.** Published sewersheds, names and outlet coordinates do not fully reconstruct underground network topology.
2. **WWTW-associated sensors are a selected population.** Results may not generalise to every upstream network CSO.
3. **Beta and lambda are estimated responses.** Their reliability varies with the number and structure of tail events, and lambda is especially unstable in some fits.
4. **Capacity is an imperfect operational proxy.** The observed cap at 1.0 suggests source semantics or censoring and does not directly measure storm-weather hydraulic headroom.

5. **Environmental variables are correlated and spatially structured.** Hydrogeology and land cover are compositional; rainfall metrics overlap strongly; company is geographically clustered.
6. **Annual dynamic rainfall remains an aggregate.** It is temporally closer to the monitoring period than climatology, but it still does not encode the rainfall preceding each individual spill.
7. **Predictive generalisation is modest.** Random/grouped holdouts are informative but do not fully guarantee transfer to a new company, region or future climate period.

11 Conclusions and next research steps

11.1 What this project established

The project created a reproducible chain from heterogeneous event records to a national WWTW-linked analytical dataset. It introduced fitted long-duration tail parameters as spatial wastewater signatures, quantified how those parameters co-vary with environmental and system characteristics, and then tested whether the complete external information set can predict unseen sites.

The final evidence supports three broad conclusions. First, long-duration CSO behaviour contains systematic spatial structure rather than being purely idiosyncratic. Second, the strongest explanatory patterns are distributed across urban-system scale, hydrogeology and wetness/rainfall rather than one single driver. Third, those static and annual-scale variables provide only modest predictive skill, implying that important hydraulic and temporal state information remains missing.

11.2 Priority next steps

The most productive extension is not an unrestricted algorithm search. Priority should be given to event-aligned rainfall and antecedent wetness, groundwater state, network/storage and pumping variables where available, improved beta-lambda reliability modelling, and transfer-aware validation across companies and space. A hierarchical event-level survival model is a natural future endpoint because it can use every long-duration event directly rather than treating fitted beta and lambda as error-free labels.

EVIDENCE NOTE The report should end on the answer, not on the methodology. Keep the final conclusion to one page in the submitted version: what was built, what was found, how strong the evidence is, and what the next decisive dataset would be.

A Data provenance and company coverage

[TO COMPLETE: Insert full source/version/year table, EIR provenance, cleaned-file paths and monitoring-period definitions.]

B Full variable dictionary

[TO COMPLETE: Insert machine-readable field names, readable names, units, source, extraction method, transformation, missingness and role.]

C Complete univariate result atlas

Use the appendix for the full beta, lambda, duration and frequency scatterplots. The main body should retain only the forest plots and a few mechanistically important examples.

D Multivariable model specifications and diagnostics

[TO COMPLETE: Insert complete M0-M4/M5 coefficient tables, VIF, sensitivity definitions, leave-one-company-out results and residual spatial diagnostics.]

E Machine-learning technical appendix

Include complete 70/15/15 split counts, hyperparameter grids, validation tables, bootstrap intervals, original-unit errors, spatial robustness, company sensitivity, permutation importance and learning-curve data. Do not repeat the main-text narrative.

F Reproducibility

Record repository commit, source hashes, Python/scikit-learn versions, random seed, authoritative master paths and figure-generation scripts.

References

References

- [1] T. Giakoumis and N. Voulvoulis. Combined sewer overflows: relating event duration monitoring data to wastewater systems' capacity in England. *Environmental Science: Water Research & Technology*, 9, 707–722, 2023.
- [2] Y. Wang, B. Huang and D. Z. Zhu. Assessment of rainfall-derived inflow and infiltration in sewer systems with machine learning approaches. *Water Science & Technology*, 89(8), 1928–1945, 2024.
- [3] J. McDonnell and D. Motta. Changing rate of urban creep and urban expansion over time and its impact upon the hydrologic response of a catchment. *Water Science & Technology*, 85(1), 383–397, 2022.
- [4] UCL CSO Spatial Drivers Project. Internal project master context, statistical analysis outputs and machine-learning release, August 2026.